

Digital Signal Processing with FAUST

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eingereicht am

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Declaration

I hereby declare and confirm that this thesis is entirely the result of my own original work. Where other sources of information have been used, they have been indicated as such and properly acknowledged. I further declare that this or similar work has not been submitted for credit elsewhere. This printed copy is identical to the submitted electronic version.

Hagenberg, June 25, 2025

Xianglei Wu

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Preface

Abstract

Here goes an abstract of the work, with a maximum of 1 page. Unlike other chapters, the abstract is usually not divided into sections and subsections. Footnotes are also not used here.

By the way, abstracts are often included in literature databases with the author and title of the work. It is, therefore, essential to ensure that the information in the abstract is coherent and complete in itself (i.e., without other parts of the work). In particular, *no literature references* are typically used at this point (as is the case also in the *title* of the thesis and the German *Kurzfassung*)! If such is needed—for example, because the paper is a further development of a particular, earlier publication—then *full* references are necessary for the abstract itself, e.g., [ZOBEL J.: *Writing for Computer Science – The Art of Effective Communication*. Springer, Singapore, 1997].

It should also be noted that special characters or list items are usually lost when records are added to a database. The same applies, of course, to the German *Kurzfassung*.

In terms of content, the abstract should not be a list of the individual chapters (the introduction chapter is intended for this purpose). However, it should provide the reader with a concise summary of your thesis. Therefore, the structure used here is necessarily different from that used in the introduction.

Kurzfassung

Dies sollte eine maximal 1-seitige Zusammenfassung Ihrer Arbeit in deutscher Sprache sein.

The German “Kurzfassung” should contain the same content as the English abstract. Therefore, try to translate the abstract precisely but not word for word. When translating, remember that certain idioms from English have no counterpart in German or must be formulated differently. Also, word order in German is very different from English (more on this in Section ??). Without knowledge of the German language, it is acceptable to resort to translators. Nevertheless, hiring a skillful person for proofreading is recommended even with the highest confidence in one’s German knowledge.

The correct translation for “diploma thesis” is *Diplomarbeit*, a “master thesis” is called *Masterarbeit*. For “bachelor’s thesis”, *Bachelorarbeit* is the appropriate translation.

By the way, for this section, the *language setting* in LaTeX should be switched from English to German to get the correct form of hyphenation. However, the correct quotation marks must be set manually (see Sections ?? and ??).

Chapter 1

Introduction

While hardware and software are important for modern audio signal processing, software has established itself as the more economical tool due to its flexibility, scalability and lower costs. Thus, we find increasingly innovative applications through findings in digital signal processing (DSP) and the advancement of modern technology, such as for sound enhancement in real-time applications for video conferencing, audio signal processing algorithms are used for noise cancellation, echo cancellation. Digital audio workstations are used in music production to process and mix recorded audio signals. They also offer a variety of audio plug-ins that virtually recreate real instruments. In view of the latest developments in deep learning, a big step has been taken in automatic speech recognition, speech separation, audio optimisation and synthesising ambient noise.

This involves converting signals, such as audio or video, from analogue to digital so that the signal data can be analysed and manipulated by the computer using mathematical algorithms.

1.1 Objectives

The aim of the thesis is to carry out a comprehensive evaluation of the FAUST programming language by illustrating its advantages and disadvantages in comparison to other programming languages in the field of audio signal processing. Properties such as performance and flexibility of FAUST are analysed and evaluated in detail. In addition, the technological possibilities offered by FAUST will be discussed.

1.2 Why FAUST?

Technological progress has led to the development of a number of specialised programming languages and architectures to enable more efficient and flexible processing of audio signals. A wide range of programming languages are now available, ranging from machine-friendly low-level languages to programmer-friendly high-level languages.

FAUST (Functional Audio Stream), on the other hand, has the advantages of both worlds at the same time. With a very intuitive programming environment and its own compiler, FAUST enables its code to be translated into an equivalent programme in languages such as C++, Java, LLVM, WebAssembly and more. Compared to the C++

code of experienced developers, the FAUST generated code can both keep up and even outperform the developers.

FAUST is characterised by its good documentation and beginner-friendliness, so that no advanced DSP knowledge is required. Nevertheless, the basic concepts of DSP make it easier to understand and use FAUST. The following chapters cover the basics of DSP and explain the features of FAUST. In addition, the functionalities, performance and technological possibilities of FAUST are explored.

Chapter 2

Basic Theories

2.1 DSP Basics

DSP is about taking information, like from visual images, sound waves or temperature, etc. from the real-world, which are being passed through an Analog-to-Digital converter as binary signals to perform mathematical operations on it to achieve various goals. E.g.: Analyzing and Filter out Noises, Enhancing the signals, Recognition and Generation of Speech, etc. With a Digital-to-Analog converter the result can be.

Signals:

Every signal has its own respective parameter that can be displayed on a graph. As it has already been said that signals are information from the real-world, they contain the characteristic of being continuous. It can be imagined as a voltage that varies its output over time. Those signal parameters (e.g., voltage and time) are being quantified during the AD-Conversion to limit the signal of both parameters to discrete values. Depending on the horizontal axis the domain can be differentiated between time domain and frequency domain. As in the case of FAUST, they are very well suited for implementing time-domain algorithms that can be easily represented as block diagrams.

Quantization:

Quantization is the process in AD-Conversion of converting a continuous range of values into a finite sequence of discrete values. The reason for that is because it enables the representation of analog signals in a digital format suitable for storage, transmission and computation.

2.2 FAUST Basics

Chapter 3

State of the art

3.1 Comparison with other technologies

3.2 Usage of FAUST in the industry

Chapter 4

Faust Demonstration

4.1

chaptersfaustdemonstration.texchaptersfaustdemonstration.tex

Chapter 5

Advanced FAUST Application

Chapter 6

Evaluation of Faust

Chapter 7

Handling References to Literature and Other Sources

Note: The title of this chapter is intentionally lengthy, so it no longer fits in the page header. For this case, a shortened text for the header and the table of contents can be specified by providing an optional argument [...] to the `\chapter` statement:

```
\chapter[References to Literature]{Adding References to Literature and Other ...}
```

7.1 General Remarks

The correct use of references is essential when writing scientific documents (see also Section 7.5). Various guidelines are available for the design of references, determined among other things by the respective subject area or guidelines from publishers and universities. This template provides a scheme that is common in engineering and scientific disciplines.¹ Technically, this part is based on the program `biber`² in combination with the LaTeX package `biblatex` [17].

Reference management consists of two elements: *citations* in the text refer to entries in the *bibliography* (or multiple bibliographies). The bibliography is a compilation of all references, typically placed at the end of the document. Each citation must have an associated, unique entry in the bibliography; each item in the bibliography must likewise be referenced in the text.

7.2 Citations

Citations can be specified in several ways. Scenarios range from citing a single reference to references with additional information, such as a page number, to citing several different references simultaneously.

¹Adaptation to other specifications is relatively easy.

²<https://ctan.org/pkg/biber>

7.2.1 The `\cite` Command

To create an entry in the bibliography and refer to it in the text, LaTeX provides a central command. For citations in the running text, use the statement

```
\cite{keys} or \cite[text]{keys}.
```

Here *keys* is a comma-separated list of one or more citation keys to identify the corresponding entries in the bibliography, and *text* can specify a supplementary text to the current citation, such as chapter or page references for books. Below are some examples:

- For further details, see [18].

```
For further details, see \cite{Kopka2003}.
```

- For further details, see [18, Ch. 3].

```
For further details, see \cite[Ch.~3]{Kopka2003}.
```

- The data in [5, pp. 274–277] appear to be outdated.

```
The data in \cite[pp.~274--277]{BurgeBurger1999} appear to be outdated.
```

- Also important are [13, 24, 37].

```
Also important are \cite{Patashnik1988,Feder2006,Duden1997}.
```

In the last example, several references are listed in a single `\cite` command. Note that the entries are sorted automatically (numerically or alphabetically). Multiple consecutive `\cite` commands should not be used for this.

7.2.2 Multiple References With Additional Texts

Attaching texts to several sources simultaneously, for example, to indicate the respective page numbers, is complex. For this purpose, the `hagenberg-thesis` package offers the additional command³

```
\mcite[text1]{key1}[text2]{key2}\dots[textN]{keyN},
```

where, for each given citation, key (*key*) an associated *text* can also be specified, for example:

- Similar results can be found in [20, Ch. 2; 10, Sec. 3.6; 11, pp. 5–7].

```
Similar results can be found in
\mcite[Ch.~2]{Loimayr2019}[Sec.~3.6]{Drake1948}[pp.~5--7]{Eberl1987}.
```

For better readability, the output—unlike the regular `\cite`—includes a *semicolon* (;) as a separator between the entries. However, sorting the entries (if desired) must be done manually; it is not done by the `\mcite` command.

³`\mcite` is defined in `hgbbib.sty` and works similar to the `\cites` command of `biblatex` (see <http://mirrors.ctan.org/macros/latex/contrib/biblatex/doc/biblatex.pdf>).

7.2.3 Suppressing Back References in the Bibliography

With the present setup, a list of the text pages on which the source was cited is automatically appended to each entry in the bibliography. In rare cases, it is helpful to suppress these back references for individual citations, by using

```
{\backtrackerfalse\cite{...}}
```

This also works with `\mcite` and other `cite` commands; note that the outer brackets are important here. For example, in item [3] of the bibliography the current page (10) should *not* be listed.

7.2.4 Common Mistakes

Several common mistakes tend to occur when working with references, especially for inexperienced authors. However, these can be easily avoided.

Placing References Outside Sentences

Citations should be placed *within* or at the *end* of a sentence (i.e., before the period), not *outside*:

Wrong: ...this is the end of the sentence. [23] And here it continues ...
 Correct: ...this is the end of the sentence [23]. And here it continues ...

References Without a Preceding Space

A citation is *always* separated from the preceding word by a space, never appended directly to the word (unlike a footnote):

Wrong: ...here goes the citation[23] and it continues ...
 Correct: ...here goes the citation [23] and it continues ...

Literal Quotes

If an entire paragraph (or more) is quoted from a source, the associated reference should be placed in the preceding text and not *within* the quote itself. As an example, the following passage from [9]:

Typographisches Design ist ein Handwerk, das erlernt werden muss. Ungeübte Autoren machen dabei oft gravierende Fehler. Fälschlicherweise glauben viele Laien, dass Textdesign vor allem eine Frage der Ästhetik ist – wenn das Schriftstück vom künstlerischen Standpunkt aus „schön“ aussieht, dann ist es schon gut „designed“. Da Schriftstücke jedoch gelesen und nicht in einem Museum aufgehängt werden, sind die leichtere Lesbarkeit und bessere Verständlichkeit wichtiger als das schöne Aussehen.

For the quote itself, the `quote` environment should be used; it demarcates the quote from one's text by employing indentations on both sides and thus reduces the risk of ambiguities (where is the end of the quote?). The above example also switches to German (see Sec. ??):⁴

⁴Note the German quotation marks inside the quote, which are set by the `smartquotes` option.

```
\begin{quote}\begin{german} quoted text ... \end{german}\end{quote}
```

If desired, the quote can also be wrapped in quotation marks *or* italicized, but not both!

Optional Extensions (Using Document Option `smartquotes`)

The `csquotes` package⁵ (automatically loaded by the `smartquotes` option) defines several additional environments for quotes, e.g.

```
\begin{displayquote} ... \end{displayquote}
```

(equivalent to `\begin{quote} ... \end{quote}`) and for foreign language quotes the environment

```
\begin{foreigndisplayquote}{language} ... \end{foreigndisplayquote}.
```

This allows, for example, a German quote⁶ *without* explicit language switching:

```
\begin{foreigndisplayquote}{german}
  quoted text ...
\end{foreigndisplayquote}
```

7.2.5 Dealing With Secondary Sources

In rare cases, it happens that one wants (or needs) to cite a source A that is not available (and thus cannot be read personally) but which is cited in *another* available source B. In this case, A is called the *original* or *primary source*, and B is called the *secondary source*. In such a scenario the following basic rules should be observed:

- Secondary sources should be *avoided* whenever possible.
- In order to quote a source in the usual form, one must always have *personally accessed* (read) it!
- Only if one can absolutely *not* get hold of the primary source, a reference via a secondary source is permissible. In this case, primary and secondary sources should be *cited together*, as shown in the example below.
- *Important*: Only the available source (B) and not the original work is included in the bibliography!

Example: Suppose one would like to quote a passage from the famous book (A) *Dialogo* by Galileo Galilei (which is difficult to obtain), referenced in a more recent work from 1969 (B). One could accomplish this, e.g., with the following footnote (all page numbers are fictional).⁷ Only the secondary source [16] is included in the actual bibliography.

⁵<https://ctan.org/pkg/csquotes>, see also section ??.

⁶Currently, only the languages `german` and `english` are defined.

⁷Galileo Galilei, *Dialogo sopra i due massimi sistemi del mondo tolemaico e copernicano*, p. 314 (1632). Quoted from [16, p. 59].

7.3 Bibliography

There are several options for creating the bibliography in LaTeX. The traditional method is to use BibTeX [24]. Another (more modern) approach uses `biber`⁸ and `biblatex`, as described below.

7.3.1 Bibliographic Data in BibTeX and BibLaTeX

BibTeX is a stand-alone program that generates a bibliography suitable for LaTeX from a “bibliographic database” (one or more text files with a given structure). Literature on using BibTeX can be found online, e.g., [24, 37].

BibTeX files can, of course, be created manually with a text editor. For many references, complete BibTeX entries are already available online. However, one should be careful because these entries are (even when provided by large institutions and publishers) *often wrong or syntactically incorrect!* Therefore, one should not adopt them unchecked and especially scrutinize the final results. Furthermore, there are specialized applications for maintaining BibTeX bibliographies, such as *JabRef*.⁹

Using biblatex and biber

This document uses `biblatex` (version 1.4 or higher) in conjunction with the software `biber`, which addresses many shortcomings of the traditional BibTeX workflow and significantly extends its capabilities.¹⁰ It adds many new entry types, which are indispensable, especially for referencing modern multimedia sources. However, this means that the bibliographic data used in `biblatex` are no longer fully backward compatible with BibTeX. It is, therefore, usually necessary to manually revise existing BibTeX data or data taken from online sources (see Section 7.3.7).

In our setup, the interface to `biblatex` is contained in the style file `hgbbib.sty`. The typical usage in the main LaTeX file is as follows:

```

1 \documentclass[master,english,smartquotes]{hgthesis}
2   ...
3 \bibliography{references}
4   ...
5 \begin{document}
6   ...
7 \MakeBibliography
8   ...
9 \end{document}
```

In the “preamble”, `\bibliography{references}`¹¹ (line 3) refers to a BibLaTeX file `references.bib`, which holds all references. If multiple BibLaTeX files are used, they can be specified in the same form.

⁸<http://mirrors.ctan.org/biblio/biber/documentation/biber.pdf>

⁹<https://www.jabref.org/>

¹⁰In fact, `biblatex` is the first radical (and long-needed) overhaul of the outdated BibTeX workflow and has already replaced the latter in many documents.

¹¹The `\bibliography` command is actually a relic from BibTeX and is replaced in `biblatex` by the `\addbibresource` statement. Both statements are equivalent, but often only `\bibliography` makes the associated `.bib` file visible in the file structure of the editor environment.

The `\MakeBibliography` statement near the end of the document (line 7) is responsible for the output of the bibliography, here with the title “References”. Two variants are possible:

`\MakeBibliography`

Creates a bibliography divided into several *categories* (see Section 7.3.2). This variant is used in the present document.

`\MakeBibliography[nosplit]`

Generates a traditional *one-piece* bibliography.

7.3.2 Reference Categories

The `hagenberg-thesis` package provides the following categories for split bibliographies (see Table 7.1):¹²

- literature – for classic publications that are available in print or online;
- avmedia – for movies, audio-visual media (on DVD, streaming, etc.);
- software – for software, APIs, computer games;
- online – for artifacts that are *only* available online.

Each reference is automatically assigned to one of these categories based on the specified BibLaTeX entry type (`@type`) (see Table 7.2). Only the most essential entry types are listed here, but they should cover most cases in practice and are explained below by examples. All entries that are not explicitly specified are assigned to the category *literature*.

7.3.3 Printed Sources (literature)

This category includes all works published in printed form, for example, in books, conference proceedings, journal articles, theses, etc. In the following examples, the BibLaTeX entry in the file `references.bib` is given, followed by the corresponding result in the bibliography.

`@book`

A single-volume book (monograph) written in its entirety by one or more authors and (typically) published by one publisher.

```
@book{BurgerBurge2022,
  author={Burger, Wilhelm and Burge, Mark James},
  title={Digital Image Processing},
  subtitle={An Algorithmic Introduction},
  publisher={Springer},
  location={Cham},
  edition={3},
```

¹²These categories are defined in the file `hgbbib.sty`. Any changes and the definition of additional categories are pretty straightforward if required.

Table 7.1: Defined reference categories and recommended BibLaTeX entry types.

<i>Literature</i> (literature)	Type	Page
Book (textbook, monograph)	@book	13
Collection (editor and multiple authors)	@incollection	14
Conference proceedings	@inproceedings	15
Article in a journal or magazine	@article	16
Thesis (bachelor, master, diploma), dissertation	@thesis	16
Technical report, lab report	@report	18
Manual, product description	@manual	18
Norm, standard	@standard	19
Laws, bills, regulations, etc.	@legislation	20
Compositions, sheet music	@book, @incollection	20
Pre-publications (e.g. conference submissions)	@unpublished	21
<i>Audiovisual media</i> (avmedia)		
Audio (CD)	@audio	22
Image, photo, graphic	@image	22
Video (on DVD, Blu-ray disk, online)	@video	23
Movie (cinema)	@movie	23
<i>Software</i> (software)		
Software product or project	@software	24
Computer game	@software	24
<i>Online sources</i> (online)		
Website, wiki entry, blog, etc.	@online	25

```

date={2022},
doi={10.1007/978-3-031-05744-1},
langid={english}
}

```

- [7] Wilhelm Burger and Mark James Burge. *Digital Image Processing. An Algorithmic Introduction*. 3rd ed. Cham: Springer, 2022. DOI: 10.1007/978-3-031-05744-1

Note: The entry field **edition** is usually only specified if there is *more* than one, especially *not for the first edition* if this is the only one! ISBNs can be safely omitted.

@incollection

A self-contained and titled contribution by one or more authors to a book or collection. **title** is the title of the contribution, **booktitle** the title of the collection, and **editor** the name of the editor.

```
@incollection{BurgeBurger1999,
```

Table 7.2: Reference categories and associated BibLaTeX entry types. In the case of a split bibliography, the entries of each category are combined in a common section. Items marked in gray are synonyms for the respective types above them.

literature	avmedia	software	online
@book	@audio	@software	@online
@incollection	@music		@electronic
@inproceedings	@video		@www
@article	@movie		
@thesis	@software		
@report			
@manual			
@standard			
@legislation			
@misc			
@unpublished			
...			

```
author={Burge, Mark and Burger, Wilhelm},
title={Ear Biometrics},
booktitle={Biometrics},
booksubtitle={Personal Identification in Networked Society},
publisher={Kluwer Academic Publishers},
date={1999},
location={Boston},
editor={Jain, Anil K. and Bolle, Ruud and Pankanti, Sharath},
chapter={13},
pages={273-285},
doi={10.1007/0-306-47044-6_13},
langid={english}
}
```

[5] Mark Burge and Wilhelm Burger. “Ear Biometrics”. In: *Biometrics. Personal Identification in Networked Society*. Ed. by Anil K. Jain, Ruud Bolle, and Sharath Pankanti. Boston: Kluwer Academic Publishers, 1999. Chap. 13, pp. 273–285. DOI: 10.1007/0-306-47044-6_13

@inproceedings

Conference paper, individual contribution in conference proceedings. Distinguish between the fields `venue` to indicate the place of the conference and `location` for the location of the publisher.

```
@inproceedings{Burger1987,
author={Burger, Wilhelm and Bhanu, Bir},
title={Qualitative Motion Understanding},
booktitle={Proceedings of the Tenth International Joint Conference on
Artificial Intelligence},
date={1987-08},
```



```

editor={McDermott, John P.},
eventdate={1987-08-23/1987-08-28},
venue={Milano},
publisher={Morgan Kaufmann Publishers},
location={San Francisco},
pages={819-821},
doi={10.1007/978-1-4615-3566-9},
langid={english}
}

```

- [6] Wilhelm Burger and Bir Bhanu. “Qualitative Motion Understanding”. In: *Proceedings of the Tenth International Joint Conference on Artificial Intelligence* (Milano, Aug. 23–28, 1987). Ed. by John P. McDermott. San Francisco: Morgan Kaufmann Publishers, Aug. 1987, pp. 819–821. DOI: 10.1007/978-1-4615-3566-9

@article

Article in a magazine, scientific journal, or daily newspaper. **volume** denotes the volume and **number** the number within that volume. The journal name (**journaltitle**) should be abbreviated only in justifiable cases to avoid misunderstandings.

```

@article{Mermin1989,
author={Mermin, Nathaniel David},
title={What's Wrong with these Equations?},
journaltitle={Physics Today},
volume={42},
number={10},
date={1989},
pages={9-11},
doi={10.1063/1.2811173},
langid={english}
}

```

- [21] Nathaniel David Mermin. “What’s Wrong with these Equations?” *Physics Today* 42.10 (1989), pp. 9–11. DOI: 10.1063/1.2811173

Note: Specifying an issue for *multiple* months (for example, in the case of a combined issue) is not possible in **biblatex** with the **month** field because it may only contain *a single* number. However, the **number** field can be used for this purpose, e.g., **number={6-7}** in [27].

@thesis

This entry type can be used for any academic thesis. The exact type is specified via the **type** attribute. The values **phdthesis**, **mathesis**, and **bathesis** indicate doctoral, master’s, and bachelor’s theses, respectively, and correctly specify the type of thesis, depending on the document language and style. Alternatively, individual content can be stored in this field.

Dissertation (Doctoral Thesis, PhD Thesis):

```
@thesis{Eberl1987,
  author={Eberl, Gerhard},
  title={Automatischer Landeanflug durch Rechnersehen},
  type={phdthesis},
  date={1987-08},
  institution={Universität der Bundeswehr, Fakultät für Raum- und
    Luftfahrttechnik},
  location={München},
  langid={ngerman}
}
```

- [11] Gerhard Eberl. “Automatischer Landeanflug durch Rechnersehen”. PhD thesis. München: Universität der Bundeswehr, Fakultät für Raum- und Luftfahrttechnik, Aug. 1987

Diploma Thesis:

Equivalent to a dissertation (see above), but with `type={Diploma thesis}`.

Master Thesis:

Equivalent to a dissertation (see above), but with `type={mathesis}`.

```
@thesis{Loimayr2019,
  author={Loimayr, Nora},
  title={Utilization of GPU-Based Smoothed Particle Hydrodynamics for
    Immersive Audiovisual Experiences},
  type={mathesis},
  date={2019-11-26},
  month={11},
  institution={University of Applied Sciences Upper Austria, Interactive Media
    },
  location={Hagenberg, Austria},
  url={https://theses.fh-hagenberg.at/thesis/Loimayr19},
  langid={english}
}
```

- [20] Nora Loimayr. “Utilization of GPU-Based Smoothed Particle Hydrodynamics for Immersive Audiovisual Experiences”. MA thesis. Hagenberg, Austria: University of Applied Sciences Upper Austria, Interactive Media, Nov. 26, 2019. URL: <https://theses.fh-hagenberg.at/thesis/Loimayr19>

The content of the `url={..}` field will be automatically typeset as a URL without any additional markup (using the `\url{..}` command).

Bachelor Thesis:

Bachelor theses are not usually considered “proper” publications, but it must still be possible to reference them if necessary, for example:

```
@thesis{Bacher2004,
  author={Bacher, Florian},
  title={Interaktionsmöglichkeiten mit Bildschirmen und großflächigen
    Projektionen},
```

```

    type={bathesis},
    date={2004-06},
    institution={University of Applied Sciences Upper Austria, Medientechnik und
        {-design}},
    location={Hagenberg, Austria},
    langid={ngerman}
}

```

- [2] Florian Bacher. “Interaktionsmöglichkeiten mit Bildschirmen und großflächigen Projektionen”. BA thesis. Hagenberg, Austria: University of Applied Sciences Upper Austria, Medientechnik und -design, June 2004

@report

These are typically numbered reports (*technical reports* or *research reports*) from companies, university institutes, or research projects. They are distinguished by the `type` attribute, which can take the values `techreport` or `resreport`. Specifying the issuing organizational unit (company, institute, faculty, etc.) and address is essential. It also makes sense to provide a corresponding URL, if available.

```

@report{Drake1948,
  author={Drake, Hubert M. and McLaughlin, Milton D. and Goodman, Harold R.},
  title={Results obtained during accelerated transonic tests of the {Bell} {XS
    -1} airplane in flights to a {MACH} number of 0.92},
  type={techreport},
  institution={NASA Dryden Flight Research Center},
  date={1948-01},
  location={Edwards, CA},
  number={NACA-RM-L8A05A},
  url={https://www.nasa.gov/centers/dryden/pdf/87528main_RM-L8A05A.pdf},
  langid={english}
}

```

- [10] Hubert M. Drake, Milton D. McLaughlin, and Harold R. Goodman. *Results obtained during accelerated transonic tests of the Bell XS-1 airplane in flights to a MACH number of 0.92*. Tech. rep. NACA-RM-L8A05A. Edwards, CA: NASA Dryden Flight Research Center, Jan. 1948. URL: https://www.nasa.gov/centers/dryden/pdf/87528main_RM-L8A05A.pdf

@manual

This publication type is suitable for technical or other documentation, such as product descriptions, manuals, presentations, white papers, etc. The documentation does not need to exist in printed form.

```

@manual{Mittelbach2023,
  author={Mittelbach, Frank and Schöpf, Rainer and Downes, Michael and Jones,
    David M. and Carlisle, David},
  title={The \texttt{amsmath} package},
  date={2023-05-13},
  version={2.17o},
  url={http://mirrors.ctan.org/macros/latex/required/amsmath/amsmath.pdf},
}

```

```
    langid={english}
}
```

- [22] Frank Mittelbach et al. *The `amsmath` package*. Version 2.17o. May 13, 2023. URL: <http://mirrors.ctan.org/macros/latex/required/amsmath/amsmath.pdf>

Often no author is named in such documents. Instead, the name of the *company* or *institution* is specified in the `author` field, but within *additional parentheses* `{..}`, so that the argument is not misinterpreted as *first name + last name*.¹³ This trick is used in the following example (among others).

@standard

References to *standards* are supported in `biblatex` through the type `@standard`. Here is a typical example:

```
@standard{WHATWGHTMLLivingStandard,
  author={{Web Hypertext Application Technology Working Group}},
  shortauthor={WHATWG},
  title={HTML},
  titleaddon={Living Standard},
  date={2023-11-06},
  url={https://html.spec.whatwg.org/multipage/},
  langid={english}
}
```

- [28] Web Hypertext Application Technology Working Group. *HTML*. Living Standard. Nov. 6, 2023. URL: <https://html.spec.whatwg.org/multipage/>

@patent

The special entry type `@patent` exists for patents, as the following example shows. `year` and `month` refer to the date of the patent grant; the specification of `holder` is optional:

```
@patent{Pike2008,
  author={Pike, Dion},
  title={Master-slave communications system and method for a network element},
  type={US Patent},
  holder={Alcatel-Lucent SAS},
  number={7,460,482},
  date={2008-12-02},
  url={https://patents.google.com/patent/US7460482},
  langid={english}
}
```

- [25] Dion Pike. “Master-slave communications system and method for a network element”. US Patent 7,460,482. Alcatel-Lucent SAS. Dec. 2, 2008. URL: <https://patents.google.com/patent/US7460482>

¹³Unlike BibTeX, `biblatex` does not accept the `organization` field as a replacement for `author` in `@manual` entries.

@legislation

Legal texts can be mapped in `biblatex` using the type `@legislation`. Since this is a non-standard type, the driver for `@misc` is used, so it is essential to specify details such as the type of publication with the `howpublished` field. The following example shows the application for a legal text (see also [4] and [12]).

```
@legislation{OoeRaumordnungsgesetz1994,
  title={Landesgesetz vom 6. Oktober 1993 über die Raumordnung im Land Oberösterreich},
  titleaddon={Öö. Raumordnungsgesetz 1994 - Öö. ROG 1994},
  howpublished={LGBL.Nr. 114/1993 zuletzt geändert durch LGBL.Nr. 111/2022},
  date={1993-12-23},
  url={https://www.ris.bka.gv.at/GeltendeFassung.wxe?Abfrage=Lr00&Gesetzesnummer=10000370},
  langid={ngerman}
}
```

- [19] *Landesgesetz vom 6. Oktober 1993 über die Raumordnung im Land Oberösterreich*. Öö. Raumordnungsgesetz 1994 - Öö. ROG 1994. LGBL.Nr. 114/1993 zuletzt geändert durch LGBL.Nr. 125/2020. Dec. 23, 1993. URL: <https://www.ris.bka.gv.at/GeltendeFassung.wxe?Abfrage=Lr00&Gesetzesnummer=10000370>

@misc

If the entry types for printed publications listed so far do not suffice, one should first look at the other types (not described in detail here) in the `biblatex` manual [17]. For example, `@collection` for a complete collection (i.e., not just a single contribution).

If none of them fit, then the `@misc` type can be used. It provides a `howpublished` text field where the type of publication can be specified individually. Likewise, the `type` field can be used to specify what kind of document is represented.

Compositions and Sheet Music

For printed compositions (sheet music or musical scores), there is, unfortunately, no particular entry type in BibLaTeX. For a *specific* edition, it is easiest to use the type `@book`, such as (see also [15, 26])

```
@book{BachBWV988,
  author={Bach, Johann Sebastian},
  title={Goldberg-Variationen für Streichquartett, BWV 988},
  editor={Anka, Dana},
  publisher={Musikverlag Hans Sikorski},
  location={Hamburg},
  date={2017},
  langid={ngerman}
}
```

- [1] Johann Sebastian Bach. *Goldberg-Variationen für Streichquartett, BWV 988*. Ed. by Dana Anka. Hamburg: Musikverlag Hans Sikorski, 2017

For compositions that have been published in a collection, one can use—as for a scientific collection—the type `@incollection`:

```
@incollection{GershwinSummertime,
  author={Gershwin, George and Heyward, DuBose},
  title={Summertime},
  booktitle={The Greatest Songs of George Gershwin},
  publisher={Chappel Music},
  location={London},
  pages={40-43},
  date={1979},
  langid={english}
}
```

- [14] George Gershwin and DuBose Heyward. “Summertime”. In: *The Greatest Songs of George Gershwin*. London: Chappel Music, 1979, pp. 40–43

@unpublished

It is becoming increasingly common for manuscripts to be published online by authors a long time before actual publication, for example, on platforms such as [arXiv.org](https://arxiv.org/)¹⁴ or [researchgate.net](https://www.researchgate.net/).¹⁵ It should be noted that releasing something online does *not formally count as a publication*, as these platforms are no publishers. Some of the works (uploaded by the authors) are *never* officially published, e.g., unaccepted submissions to conferences. When citing this reference it is crucial to determine whether the paper was accepted and actually published:

- a) *The work was indeed published*: Here the corresponding original publication must be searched and used! The citation is done in conventional form (for example, with **@inproceedings** in case of a conference volume), but *without* a reference to the online publication.
- b) *The work was not published*: If indeed no publication (or an associated *technical report*, see above) can be found, the **@unpublished** tag can be helpful:

```
@unpublished{Dai2016,
  author={Dai, Jifeng and Li,Yi and He, Kaiming and Sun, Jian},
  title={{R-FCN:} Object Detection via Region-Based Fully Convolutional
    Networks},
  date={2016},
  pubstate={prepublished},
  doi={10.48550/arXiv.1605.06409},
  langid={english}
}
```

- [8] Jifeng Dai et al. “R-FCN: Object Detection via Region-Based Fully Convolutional Networks”. 2016. DOI: 10.48550/arXiv.1605.06409. Pre-published

In the latter case, the specification of the link (via **doi** or alternatively **url**) is essential. Details on the **pubstate** field (**prepublished**) can be found in the **biblatex** documentation [17, Sec. 4.9.2.11]. Otherwise (if unknown or not published),

¹⁴<https://arxiv.org/>

¹⁵<https://www.researchgate.net/>

`note={unpublished}` or `note={unveröffentlicht}`

can be used instead of `pubstate`.

7.3.4 Movies and Audio-Visual Media (avmedia)

This category covers audio-visual productions such as movies, sound recordings, CDs, DVDs, and VHS tapes, thus meaning works published in physical (but not printed) form. It does not include audio-visual creations (sound recordings, images, videos) that are exclusively available online—these should be referenced with the entry type `@online` (see Table 7.2 and Section 7.3.6).

The `@audio`, `@video`, and `@movie` types described below are *not* BibTeX standard types. However, they are represented in `biblatex` (implicitly replaced by `@misc`) and are recommended here to allow the automatic assignment of items in the bibliography.

`@audio`

Here is an example of the specification of an audio CD:

```
@audio{Zappa1995,
  author={Zappa, Frank},
  title={Freak Out!},
  type={audiocd},
  date={1995-05},
  organization={Rykodisc, New York},
  langid={english}
}
```

[34] Frank Zappa. *Freak Out!* Audio CD. Rykodisc, New York, May 1995

Instead of `type={audiocd}` one could also use `howpublished={Audio CD}`.

`@image`

The following example shows the reference to a digitally available photo, which is also used in Figure ??:

```
@image{CocaCola1940,
  author={Wolcott, Marion Post},
  title={Natchez, Miss.},
  note={Library of Congress Prints and Photographs Division Washington, Farm
    Security Administration/Office of War Information Color Photographs},
  date={1940-08},
  url={https://www.loc.gov/pictures/item/2017877479/},
  langid={english}
}
```

[33] Marion Post Wolcott. *Natchez, Miss.* Library of Congress Prints and Photographs Division Washington, Farm Security Administration/Office of War Information Color Photographs. Aug. 1940. URL: <https://www.loc.gov/pictures/item/2017877479/>

@video

The example below shows a link to a YouTube video:

```
@video{HistoryOfComputers2008,
  title={History of Computers},
  date={2008-09-24},
  url={https://www.youtube.com/watch?v=LvKxJ3bQRKE},
  langid={english}
}
```

- [30] *History of Computers*. Sept. 24, 2008. URL: <https://www.youtube.com/watch?v=LvKxJ3bQRKE>

Here is an example of referencing a DVD edition:

```
@video{Futurama1999,
  author={Groening, Matt},
  title={Futurama},
  titleaddon={Season 1 Collection},
  howpublished={DVD},
  date={2002-02},
  organization={Twentieth Century Fox Home Entertainment},
  langid={english}
}
```

- [29] Matt Groening. *Futurama*. Season 1 Collection. DVD. Twentieth Century Fox Home Entertainment, Feb. 2002

In this case, the specified date is the *date of issue*. If no unique author can be named, omit the **author** field and wrap the corresponding information in the **note** field, as shown in the example below.

@movie

This entry type is intended for movies. No author is specified because it is usually not possible to name a specific author for a film production. In the following example (see also [32]) the relevant data are provided in the **note** field:¹⁶

```
@movie{Nosferatu1922,
  title={Nosferatu -- A Symphony of Horrors},
  howpublished={Film},
  date={1922},
  note={Drehbuch/Regie: F.\ W.\ Murnau. Mit Max Schreck, Gustav von Wangenheim, Greta Schröder.},
  langid={english}
}
```

- [31] *Nosferatu – A Symphony of Horrors*. Film. Drehbuch/Regie: F. W. Murnau. Mit Max Schreck, Gustav von Wangenheim, Greta Schröder. 1922

The specification `howpublished={Film}` makes sense here to avoid confusion with a possible book of the same name.

¹⁶By the way, *biblatex* nicely ensures that the dot at the end of the **note** text is not duplicated in the output.

Time Codes for Music Recordings and Movies

A reference to a specific passage in a music or movie can be accomplished similarly to the page reference in a printed work. Particularly legendary (and often parodied), for example, is the shower scene in *Psycho* [32, T=00:32:10]:

```
\cite[T=00:32:10]{Psycho1960}
```

As an alternative to the simple timecode “T=hh:mm:ss”, it is possible to specify the position of a specific frame by using the corresponding *timecode* “TC=hh:mm:ss:ff”, e.g., [32, TC=00:32:10:12] for frame ff=12:

```
\cite[TC=00:32:10:12]{Psycho1960}
```

7.3.5 Software (@software)

This entry type is especially suitable for computer games (in the absence of a particular entry type).

```
@software{LegendOfZelda1998,
  author={Miyamoto, Shigeru and Aonuma, Eiji and Koizumi, Yoshiaki},
  title={The Legend of Zelda: Ocarina of Time},
  howpublished={N64 Cartridge},
  publisher={Nintendo},
  date={1998-11},
  langid={english}
}
```

- [35] Shigeru Miyamoto, Eiji Aonuma, and Yoshiaki Koizumi. *The Legend of Zelda: Ocarina of Time*. N64 Cartridge. Nov. 1998

The following is an example of the reference to a typical software project:

```
@software{SpringFramework,
  title={Spring Framework},
  url={https://github.com/spring-projects/spring-framework},
  langid={english}
}
```

- [36] *Spring Framework*. URL: <https://github.com/spring-projects/spring-framework>

7.3.6 Online Sources (@online)

In the case of references to online resources, three cases need to be distinguished:

- A. Referring to a web page in general, such as the “Panasonic products for business” page.¹⁷ In this case, there is no reference to a specific “work”; therefore, it is *not included* in the bibliography. Instead, as shown in the previous sentence, it is sufficient to use a simple footnote with `\footnote{\url{...}}`.

¹⁷<http://business.panasonic.co.uk/>

- B. Some piece of printed or audio-visual work (see Secs. 7.3.3 and 7.3.4) is *additionally* available online. In this case, however, the primary publication is *not* “online”, and it is sufficient to specify the associated link in the `url` field, which can be specified for any entry type.
- C. A publication in the broadest sense that is *exclusively* available online, such as a wiki or blog entry. The *online* category is precisely (and *only*) intended for this type of source.

Example: Wiki Entry

Due to the volume and increasing quality of these entries, their inclusion in a bibliography seems justified. For example, a *relic shrine* (German: “Reliquienschein”) is a casket in which the relics of one or more saints are kept [38].

```
@online{WikiReliquienschein2023,
  title={Reliquienschein},
  url={https://de.wikipedia.org/wiki/Reliquienschein},
  date={2023-09-22},
  urldate={2023-11-06},
  langid={ngerman}
}
```

[38] *Reliquienschein*. Sept. 22, 2023. URL: <https://de.wikipedia.org/wiki/Reliquienschein> (visited on 11/06/2023)

In this case, the reference consists mainly of a URL. With `date`, one can specify the current version at that time. The (optional) specification of `urldate` (in YYYY-MM-DD format) automatically inserts the information when the online document was accessed.

Technically, only the `url` field is required for online sources, but specifying further details (e.g., `author`) is possible. However, if *no* author is available, then—as shown in the examples above—at least a meaningful *title* (`title`) should be added, which may be used for sorting the bibliography.

7.3.7 Tips for Creating BibLaTeX Files

The following items should be considered for creating correct BibLaTeX files.

date Attribute

While in classic BibTeX, the year and month of a publication are specified using the attributes `year` and `month`, for pure BibLaTeX bibliographies (as in this document), the attribute `date` is better suited. Entries are made in the format YYYY-MM-DD, which can also consist of only the year (YYYY) or year and month (YYYY-MM). Likewise, durations can also be defined in the format YYYY-MM-DD/YYYY-MM-DD. Related fields are `origdate` (original publication date for, e.g., a reprint or translation), `eventdate` (date of a conference), and `urldate` (access date of an URL).

Should `year` and `month` nevertheless be used, note that the latter is numeric in BibLaTeX (in contrast to BibTeX) and is specified, for example, in the form `month={8}` (for August).

langid Attribute

The `langid` attribute enables the correct typesetting of multilingual bibliographies. If possible, it should be specified for every reference, e.g.,

```
langid={english} or langid={ngerman}
```

for an English- or German-language source, respectively.

edition Attribute

The numeric `edition` field is used to specify the edition of a reference. Only the number itself has to be specified, e.g., `edition={3}` for a third edition. The complete text is added automatically depending on the language setting (e.g., “3rd edition” or “3. Auflage”). As already described on page 13 (under `@book`), in case of a *first* edition, the `edition` field should *not* be specified if there is no other edition!

Caution When Incorporating Existing Bibtex Entries

Many publishers and literature brokers offer ready-made BibTeX entries for download. However, great care should be taken because these entries are often incomplete, inconsistent, or syntactically incorrect! They should *always* be checked for correctness when importing them! Particular attention should be paid to the correct specification of the first names (*fn*) and last names (*ln*), preferably in the form¹⁸

```
author={ln1, fn1a fn1b and ln2, fn2a ...}.
```

This is especially important for multi-part surnames because otherwise, the first and last names cannot be correctly assigned, e.g.,

```
author={van Beethoven, Ludwig and ter Linden, Jaap}
```

for a (fictional) piece of joint work by *Ludwig van Beethoven* and *Jaap ter Linden*.

Omissions or errors are often found in entries of `volume`, `number`, and `pages`, especially in collections (`@incollection`) and conference proceedings (`@inproceedings`). Also, the names of conferences and meeting venues are often not specified correctly (even in official ACM and IEEE sources). ISBN and ISSN are usually redundant and should be omitted. However, a DOI (Digital Object Identifier) entry is useful. This unique number is assigned a hyperlink by BibLaTeX that points to the source of the work (usually that of the publisher). To avoid duplicate hyperlinks in the entry, the `hagenberg-thesis` package automatically removes the `url` field if a DOI is present.

Since imported entries are almost always in BibTeX and not BibLaTeX notation, they should be adjusted to correct types if necessary and to make use of the current detail fields.

Common Citation Errors

Check the finished bibliography carefully for *completeness* and *consistency*. For each reference, is it clear how and where it was published? Are the details sufficient to locate the source? Here is a list of the most critical provisions related to the bibliography:

¹⁸Note how the commas are placed! The keyword `and` separates the names of the individual authors.

- Check all entries for missing or misinterpreted items!
- Check all authors' names and first names, are the abbreviations (of first names) consistent?
- Check the capitalization and punctuation of all entries and correct them if necessary.
- Books: Check the publisher's name and the place of publication for completeness, consistency, and redundancies.
- *Omit* URLs unless they are indispensable! This is especially true for books and conference papers. Instead, provide a DOI if available—it will also be linked to the source in the PDF.
- Journal articles: Always specify the full name of the journal, e.g., “ACM Transactions on Computer-Human Interaction” instead of “ACM Trans. Comput.-Hum. Interact.”! Do not forget the page numbers!
- Conference proceedings: Refer to conference proceedings consistently in the form “Proceedings of the *XY Conference on Something ...*”. Indicate the conference venue, and remember to specify the page numbers!
- For technical reports, master theses, and dissertations, the institution (university and department, company, etc.) *must* be provided!

Listing of All References

Using the `\nocite{*}` statement—placed anywhere in the document—will list *all* existing entries of the BibLaTeX file in the bibliography, including those for which there is no explicit `\cite{}` call in the text. This is useful for getting an overview of all references while writing the thesis. Normally, however, all cited sources must also be referenced in the text!

7.4 Using the APA Citation Style

As an alternative to the numeric citation style set in this document (`numeric-comp`), the style of the American Psychological Association¹⁹ (APA) may be used. This should be in accordance with the institute's guidelines and agreed upon with the supervisor. This type of reference formatting uses the author's name and the year of publication instead of a number in square brackets. The formatting of the entries in the bibliography is also different.

To use APA as the citation style throughout the document, the document option “`apa`” must be specified in the main file, e.g.,

```
\documentclass[master,english,smartquotes,apa]{hgbthesis}
```

Depending on the type of use, several *different commands* are necessary to reference sources in the text, as described below.

¹⁹<https://apastyle.apa.org/style-grammar-guidelines/references/>

7.4.1 Narrative Citations

For narrative citations, the reference is used as the subject or object of the sentence. The year is placed after the author's name in parentheses. To create this kind of citation use

```
\textcite{keys}.
```

Example:

```
\textcite{Daniel2018} give a brief introduction to \latex, whereas  
\textcite{Oetiker2021, Kopka2003} go into more detail.
```

Daniel et al. (2018) give a brief introduction to LaTeX, whereas Kopka and Daly (2003) and Oetiker et al. (2021) go into more detail.

7.4.2 Narrative Citations Within Parentheses

If a reference is to be used *within* parentheses, these must be omitted from the citation. The associated command is

```
\nptextcite{keys}.
```

Example:

```
In any case, it is recommended to obtain literature on the topic of \latex  
(\eg, \nptextcite{Daniel2018, Oetiker2021, Kopka2003}).
```

In any case, it is recommended to obtain literature on the topic of LaTeX (e.g., Daniel et al., 2018; Kopka & Daly, 2003; Oetiker et al., 2021).

7.4.3 Parenthetical Citations

Parenthetical citations are used when the reference should be cited at the end of a sentence or statement. The author's name and year are enclosed in parentheses and separated by a comma. In this case, use

```
\parencite{keys}.
```

Example:

```
For \latex, there are short introductions \parencite{Daniel2018}, as well as  
more comprehensive works \parencite{Oetiker2021, Kopka2003}.
```

For LaTeX, there are short introductions (Daniel et al., 2018), as well as more comprehensive works (Kopka & Daly, 2003; Oetiker et al., 2021).

Appearance in the Bibliography

The references used above are displayed in the bibliography as follows:

- Daniel, M., Gundlach, P., Schmidt, W., Knappen, J., Partl, H. & Hyna, I. (2018, April 8). *L^AT_EX2e-Kurzbeschreibung*. Version 3.0c. <http://mirrors.ctan.org/info/lshort/german/l2kurz.pdf>. (cit. on pp. 63, 64)
- Kopka, H. & Daly, P. W. (2003). *Guide to L^AT_EX* (4th ed.). Addison-Wesley. (Cit. on pp. 63, 64).
- Oetiker, T., Partl, H., Hyna, I. & Schlegl, E. (2021, March 9). *The Not So Short Introduction to L^AT_EX2e: Or L^AT_EX2e in 139 minutes*. Version 6.4. <http://mirrors.ctan.org/info/lshort/english/lshort.pdf> (cit. on pp. 63, 64).

7.5 Plagiarism and Paraphrasing

Plagiarism means representing somebody else's work as one's own creation, in part or whole, whether consciously or unconsciously. Plagiarism is not a new problem in higher education but has increased dramatically in recent years due to the widespread availability of electronic sources. Many universities now also use electronic resources (some of which are not accessible to students) as a countermeasure. Therefore, one should expect any submitted work to be routinely checked for plagiarism!

Plagiarism is by no means considered a trivial offense. Even if discovered at some later point, this can, in the worst case, lead to the subsequent (and final) revocation of the academic degree. To avoid such problems, one should better be overcautious and at least observe the following rules:

- Adopting short text passages is only permitted with correct citation, whereby the text quotation's scope (beginning and end) must be clear in each case.
- In particular, it is not acceptable to mention a reference only at the beginning and then repeatedly adopt non-cited text passages as one's own creation.
- Under no circumstances tolerable is to directly adopt or *paraphrase* longer text passages, whether with or without citation. Passages used indirectly or translated from another language must also be cited properly!

In case of doubt, more detailed rules can be found in any good book on scientific writing. To be safe, asking your thesis advisor is never a mistake.

Chapter 8

Closing Remarks¹

This should be a summary of your thesis, which may also address the process of its creation, experiences, insights and problems encountered during the implementation (but no personal issues), areas for improvement, possible extensions, etc. Was the topic well chosen, what was eventually achieved, what points remain open and how could work continue from here?

8.1 Read and Let Read

When your thesis is finished, the first thing you should do is to read it over again *completely* and *carefully* yourself, even though you might not feel inclined to once more look at something you have worked on for so long. In addition, it is highly recommended to have another person do this as well—you will be amazed at how many additional mistakes you had missed.

The use of AI-assisted writing assistants such as *Grammarly*² or *LanguageTool*³ can also be quite useful. However, the suggestions from these tools should not simply be accepted blindly but with caution.

8.2 Checklist

Finally, Table 8.1 gives a brief checklist of important items that most frequently are the cause of errors. If an official thesis review is required at your university, such and similar items are typically checked by the assigned *thesis editor* as well.

¹This note only demonstrates the (rarely necessary) use of footnotes in headings. See the source text for how this is done. Make sure the footnote does not appear in the table of contents as well!

²<https://grammarly.com/>

³<https://languagetool.org/>

Table 8.1: List of important items as typically checked during an academic *thesis review*.

- | |
|--|
| <ul style="list-style-type: none"> ☐ Title page: length of title (line breaks), name, program of study, date. ☐ Declaration: complete signature. ☐ Table of contents: balanced structure, depth, length of headings. ☐ Abstract/Kurzfassung: precise summary, appropriate length, same content and structure. ☐ Chapter/section titles: length, style, clarity. ☐ Layout/typography: clean printout (no raster fonts), no “manual” spacing between paragraphs or indentations, no overlong lines, highlighting, font size, footnote placement. ☐ Language: gender-appropriate wording (no generic masculine or general clause), objective, factual wording. ☐ Punctuation: hyphens and dashes placed correctly, proper spacing after periods (especially after abbreviations), correct (front/back) quotation marks. ☐ Figures: quality of graphics and images, font size and type in figures, proper placement of figures and tables, captions. Are <i>all</i> figures (tables) referenced in the text? ☐ Equations/formulas: placement of mathematical elements in continuous text, correct use of displayed equations and mathematical symbols. ☐ References: citations properly referenced, including page and chapter references; no unresolved cross references (??). ☐ Bibliography: type of publication must be clear in all cases, consistent and complete entries, online sources (URLs) cleanly cited. ☐ Other: contents of appendix, PDF paper size ($A4 = 8.27 \times 11.69$ inches), print size and quality. |
|--|

Appendix A

Technical Information

A.1 Current Package Version

Date	File
2023/11/06	<code>hgb.sty</code>

A.2 Additional Details

This package is designed for UTF-8 encoded source files and supports LaTeX in direct PDF mode only.¹

A.2.1 Technical Requirements

A current LaTeX installation including

- `biber` (modern replacement for BibTeX, Version ≥ 1.5),
- `biblatex` package (version ≥ 2.5 , 2013/01/10),
- Latin Modern fonts (package `lmodern`).²

In addition, a text editor for UTF-8 encoded (Unicode) files, as well as software for opening and viewing PDF files.

A.2.2 Use Under Windows

A typical installation under Windows looks like this:

1. **MikTeX**³ (basic LaTeX environment),

¹The “classic” DVI-PS-PDF process is no longer supported.

²<https://ctan.org/pkg/lm>, <https://tug.org/FontCatalogue/latinmodernroman/>

³<https://miktex.org/> – Generally, the *complete installation* of MikTeX (“Complete MiKTeX”) is recommended, as it already contains all necessary additional packages and font files. During installation, make sure that the automatic installation of required packages is enabled by “*Install missing packages on-the-fly: = Yes*” (NOT “*Ask me first*”)! It is also recommended to update the installed packages immediately after installing MikTeX and periodically thereafter using the **MikTeX Console** program.

2. **TeXstudio**⁴ (text editor, supports UTF-8 and includes an integrated PDF viewer).

Alternative editors and PDF viewers are:

1. TeXnicCenter,⁵
2. Texmaker,⁶
3. Lyx,⁷
4. TeXworks,⁸
5. WinEdt,⁹
6. Sumatra PDF (“LaTeX friendly” PDF viewer).¹⁰

A.2.3 Use Under macOS

For macOS, the following configuration is recommended:

1. **MacTeX**¹¹ (basic LaTeX environment),
2. **TeXstudio** (text editor, supports UTF-8 and includes an integrated PDF viewer).

Alternative editors and PDF viewers are:

1. Texmaker,
2. Lyx,
3. TeXworks,
4. Skim (“LaTeX friendly” PDF viewer).¹²

A.2.4 Use Under Linux

Under Linux the following setup can be used:

1. **TeX Live**¹³ (basic LaTeX environment),

⁴<https://www.texstudio.org/>

⁵<https://www.texniccenter.org/>

⁶<https://www.xmlmath.net/texmaker/>

⁷<https://www.lyx.org/>

⁸<https://www.tug.org/texworks/>

⁹<https://www.winedt.com/>

¹⁰<https://www.sumatrapdfreader.org/>

¹¹<https://tug.org/mactex/> – Current MacTeX distributions usually require a mostly up-to-date version of macOS. On older versions, *TeXLive* can alternatively be installed with a special installation script. To keep the packages of the LaTeX distribution up-to-date, the *TeX Live Utility* program should be run regularly.

¹²<https://skim-app.sourceforge.io/>

¹³<https://tug.org/texlive/> – An installation under Linux is—depending on the distribution used—most easily done with the help of the associated package management system (e.g., `apt-get`).

2. **TeXstudio** (text editor, supports UTF-8 and includes an integrated PDF viewer).

Alternative editors and PDF viewers are:

1. Texmaker,
2. Lyx,
3. TeXworks,
4. qpdfview (“LaTeX friendly” PDF viewer).¹⁴

A.2.5 Using Online LaTeX Environments

Besides using a local LaTeX installation and editor, there are now also good online environments that allow to create LaTeX documents directly in the browser. The LaTeX environment is installed on the servers of the provider. Documents can be created in the online editor or existing templates (such as this document) uploaded and edited. Most platforms also allow collaborative work on the same document.

When using such environments, it is highly recommended to perform regular *backups* of your online data while working, so that in the worst case you don’t have to start all over again.

Overleaf

The most popular editor (tested with this template) is *Overleaf*¹⁵. To quickly import template documents from the `hagenberg-thesis` package, the import links in the *readme* section of this project’s Github repository¹⁶ can be used directly.

Other Online Services

Besides, there are other online environments for LaTeX and their number is constantly growing, for example:

1. Papeeria,¹⁷
2. CoCalc.¹⁸

¹⁴<https://launchpad.net/qpdfview>

¹⁵<https://www.overleaf.com/>

¹⁶<https://github.com/Digital-Media/HagenbergThesis>

¹⁷<https://papeeria.com/>

¹⁸<https://cocalc.com/>

Appendix B

Supplementary Materials

This is a listing of supplemental materials to this thesis that have been submitted to the university for digital archiving (e.g., as a ZIP file). This should only serve as an example, you can adapt the structure to your own needs!

B.1 PDF Files

Path: /

thesis.pdf final thesis (complete document)

B.2 Media Files

Path: /media

*.ai, *.pdf illustrations (line graphics)
*.jpg, *.png pictures (raster images)
*.mp3 audio files
*.mp4 video files

B.3 Copies of Online Sources

Path: /online-sources

Reliquienschrein-Wikipedia.pdf [38]

Appendix C

Questionnaire

This section demonstrates—as an example—how to include an external PDF document in your own LaTeX manuscript. This problem arises relatively often, e.g., in connection with questionnaires that one has created and/or used in the thesis. Therefore, exactly this case is shown here.¹ It is important that the *page formatting* of the own document is not disrupted and the *page numbering* correctly considers the inserted pages.

C.1 The pdfpages Package

The LaTeX package `pdfpages`² is (currently) the only option for this purpose. It is loaded automatically by `hgb.sty` with

```
\RequirePackage{pdfpages}
```

The included PDF document (a two-page questionnaire) is located in `images/fragebogen.pdf`. To include all (2) pages of the PDF file in the current document, we use the instruction

```
\includepdf[pages=1-,width=\textwidth,frame=true,pagecommand={}]{images/fragebogen}
```

The included pages are automatically scaled to the text width of the LaTeX document by `width=\textwidth` and `frame=true` adds a surrounding border.

This example assumes that the external PDF document is in A4 page format. With other formats you may have to adjust the scaling “manually” if the pages become too tall (e.g., with `width=0.9\textwidth`).

It is also important that all *fonts* used in the external PDF document are correct and fully *embedded*, otherwise the PDF document generated by LaTeX may not be viewable in another system environment!

C.2 References to Included PDF Pages

If you want to refer to specific pages in the included PDF, the easiest way is to import single pages one by one and add a *label* to each, as in this example:

¹With a nice questionnaire of the *Upper Austrian Energy Saving Association* (<https://www.energiesparverband.at/>).

²<https://ctan.org/pkg/pdfpages>

```
\includepdf[pages=1,width=\textwidth,frame=true,  
pagecommand={\label{PDF1}}]{images/fragebogen}  
\includepdf[pages=2,width=\textwidth,frame=true,  
pagecommand={\label{PDF2}}]{images/fragebogen}
```

For example, in this case you could use `\pageref{PDF2}` to specify the current page number of the 2nd page of the included PDF document. Many other options (e.g., specifying page intervals) can be found in the detailed documentation for the `pdfpages` package.

Fragebogen Ein- & Zweifamilienhaus, Wohnung		www.energiespargemeinde.at		Energiespar Gemeinde
Bei gemeinsam versorgten Haushalten (eine Energierechnung) bitte übergreifend ausfüllen!				
PLZ: _____	PERSONEN IM HAUSHALT _____	☐ Einfamilienhaus ☐ Zweifamilienhaus ☐ Wohnung		
GEBÄUDE				
• beheizte Wohnfläche	_____ m ²			
• Baujahr	☐ vor 1919 ☐ 1919 - 1944 ☐ 1945 - 1960 ☐ 1961 - 1980 ☐ 1981 - 2000 ☐ nach 2000			
• Falls Zubau, wann?	_____			
HAUSTYP				
• Dachform	☐ Satteldach ☐ Flachdach ☐ Giebeldach ☐ Schopfwalmdach ☐ Pultdach ☐ Sonstiges: _____			
• Anzahl der beheizten Stockwerke	☐ 1 ☐ 2 ☐ 3			
• durchschn. Raumhöhe	_____ m			
• Keller	☐ ohne Keller ☐ unbeheizter Keller ☐ beheizter Keller			
WARMWASSERBEREITUNG				
• System	☐ Warmwasser mit der Heizung ☐ Strom ☐ Wärmepumpe			
HEIZUNG				
• Energieträger bzw. Heizungstyp	☐ Heizöl ☐ Nahwärme ☐ Pellets ☐ Wärmepumpe ☐ Hackgut ☐ Scheitholz ☐ Erdgas ☐ Flüssiggas ☐ Kohle, Koks ☐ Stromheizung ☐ Sonstiges: _____			
• Energieverbrauch pro Jahr	_____			
	☐ Liter ☐ kWh ☐ kg ☐ m ☐ m ³ ☐ € ☐ srm ☐ fm			
• Baujahr Heizung	☐ vor 1978 ☐ zw. 1978 und 1994 ☐ nach 1994 ☐ neue Heizung (bis 3 Jahre alt)			
AUSSENWAND				
• Wanddicke (ohne Dämmung)	_____ cm			
• Baumaterial	_____			
• Dämmstärke	_____ cm			
FENSTER				
• Fenstertyp	☐ Einscheiben Verglasung (bis 1950) ☐ Verbundfenster (1960 - 1980) ☐ Kastenfenster (bis 1960) ☐ Passivhausfenster (2003 - jetzt) ☐ Isolierglasfenster mit 2 Scheiben (1975 - 1995) ☐ Isolierglasfenster mit 3 Scheiben (1975 - 1995) ☐ Wärmeschutzfenster mit 2 Scheiben (1995 - jetzt) ☐ Wärmeschutzfenster mit 3 Scheiben (2000 - jetzt) ☐ Sonstiges: _____			
ZUSATZHEIZUNG ODER 2TER ENERGIETRÄGER				
• Zusätzlicher Energieträger bzw. Heizungstyp	_____ (z.B.: Pellets, Erdgas, Holz, Nahwärme, Kohle, etc.)			
• Energieverbrauch pro Jahr	_____			
	☐ Liter ☐ kWh ☐ kg ☐ m ☐ m ³ ☐ € ☐ srm ☐ fm			
MOBILITÄT				
	Privatfahrzeug 1	Privatfahrzeug 2		
• Treibstoff	_____	_____		
• Gesamtkilometer pro Jahr	_____	_____	km	
• Verbrauch pro 100 km	_____	_____	Liter	
• Einfache Wegstrecke (z.B. zur Arbeit)	_____	_____	km	
STROMVERBRAUCH PRO JAHR				
• Summe	_____	☐ kWh ☐ Euro		
ERNEUERBARE ENERGIEN				
• Ich besitze	_____ m ² Sonnenkollektoren			
	_____ kWp Photovoltaik			
	_____ kW Wasserkraft			
LETZTE SANIERUNGSMASSNAHMEN (JAHR)				
• Außenwand	_____	• Oberste Geschosdecke	_____	
• Fenster	_____			
NUR FÜR WOHNUNG				
• Anzahl der Außenwände	☐ 1 ☐ 2 ☐ 3 ☐ 4			
• Oberhalb der Wohnung	☐ Wohnung ☐ Dach ☐ Dachboden			
• Unterhalb der Wohnung	☐ Wohnung ☐ Keller ☐ erdanliegender Boden			
OBERSTE GESCHOSSECKE				
• Material bzw. Konstruktionstyp	_____ (z.B.: Betondecke, Ziegeldecke, Zangendecke, etc.)			
• Dämmstärke	_____ cm			
KELLERDECKE				
• Material bzw. Konstruktionstyp	_____ (z.B.: Betondecke, Ziegeldecke, Hohlkörper mit Beschüttung, etc.)			
• Dämmstärke	_____ cm			
HAUSTECHNIK				
• Warmwasserleitungen wärmedämmte?	☐ Ja ☐ Nein			
• Heizungsleitungen wärmedämmte?	☐ Ja ☐ Nein			
• Wärmeabgabe	☐ Fußboden oder Wandheizung ☐ Heizkörper ☐ Heizkörper & Fußboden-Wandheizung			
• Regelung	☐ Händisch ☐ Außentemperaturgeführt, Raumthermostat ☐ Außentemperaturgeführt ☐ Nur Thermostatventil			
• Raumtemperatur in der Heizperiode	_____ °C			
• Lüftung mit Wärmerückgewinnung	☐ Ja ☐ Nein			

Fragebogen		www.energiespargemeinde.at		Energiespar Gemeinde																																																	
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Wenn Sie die Erhebung nicht online beantwortet haben, sondern mittels dieses Fragebogens, so ersuchen wir Sie, diesen an folgende Adresse zu senden oder direkt im Gemeindeamt abzugeben. Danke für Ihre Mitarbeit!																																																					

Appendix D

LaTeX-Quellcode

Main File (main.tex)

Note: This should just be an *example* of how to include source code in the document's Appendix. It is accomplished with the following instructions:

```
\begin{footnotesize}
  \verbatiminput{main.tex}
\end{footnotesize}
```

Of course, the LaTeX source code of one's thesis is usually *not* interesting enough to be reproduced here!

```
%%% File encoding: UTF-8
%%% äöüÄÖÜß <-- no German umlauts here? Use an UTF-8 compatible editor!

%%% Magic comments for setting the correct parameters in compatible IDEs
% !TeX encoding = utf8
% !TeX program = pdflatex
% !TeX spellcheck = en_US
% !BIB program = biber

\RequirePackage[utf8]{inputenc} % Remove when using lualatex or xelatex!
\RequirePackage{hgbpdfa} % Creates a PDF/A-2b compliant document

\documentclass[bachelor,english,smartquotes]{hgbthesis}
% Valid options in [..]:
%   Type of work: 'diploma', 'master' (default), 'bachelor', 'internship'
%   Additionally for a thesis exposé: 'proposal' (for 'bachelor' and 'master')
%   Main language: 'german' (default), 'english'
%   Turn on smart quote handling: 'smartquotes'
%   APA bibliography style: 'apa'
%%%-----

\graphicspath{{images/}} % Location of images and graphics
\logofile{logo} % Logo file: images/logo.pdf (no logo: \logofile{})
\bibliography{references} % BibLaTeX bibliography file (references.bib)

%%%-----
\begin{document}
%%%-----
```

```

%%%-----
% Title page entries
%%%-----

\title{Digital Signal Processing with FAUST}
\author{Xianglei Wu}
\programname{Software Engineering}

%\programtype{Fachhochschul-Bachelorstudiengang} % select/edit
\programtype{Fachhochschul-Bachelorstudiengang}

\placeofstudy{Hagenberg}
\dateofsubmission{2025}{06}{25} % {YYYY}{MM}{DD}

\advisor{Erik Pitzer, FH-Prof. DI (FH) Dr.} % optional

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%%%-----
\frontmatter % Front part (roman page numbers)
%%%-----

\maketitle
\tableofcontents

\include{front/preface} % A preface is optional
\include{front/abstract}
\include{front/kurzfassung}

%%%-----
\mainmatter % Main part (arabic page numbers)
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\include{chapters/introduction}
\include{chapters/theoryandbasics}
\include{chapters/stateoftheart}
\include{chapters/faustdemonstration}
\include{chapters/advancedfaustapplication}
\include{chapters/evaluationoffaust}
\include{chapters/literature}
\include{chapters/closing}

%%%-----
\appendix % Appendix
%%%-----

\include{back/appendix_a} % Technical supplements
\include{back/appendix_b} % Contents for electronic submission
\include{back/appendix_c} % Included other PDF document
\include{back/appendix_d} % Source text of this document

%%%-----
\backmatter % Back part (bibliography, glossary, etc.)
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\MakeBibliography % References

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% Special page for checking print size  
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\include{back/printbox}  
  
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Software

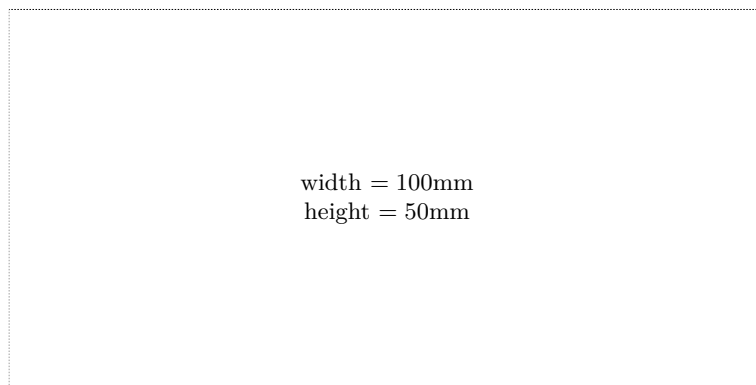
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